

- Q.30 Describe the various protection methods with sketches used in excavation works?
- Q.31 Differentiate between Fall Arrest Systems and Fall Restraint system along with suitable examples.
- Q.32 Explain with the help of diagram the minimum height required for arresting a fall effectively using FBH (Height of the worker, Lifeline height above working platform etc. may be assumed and mentioned clearly)
- Full body harness with shock absorber
 - Full body Harness without shock absorber
- Q.33 Explain various safety devices used in Cranes used for lifting operations.
- Q.34 What do you mean by electrical safety devices? Differentiate between RCCB over ELCB?
- Q.35 Describe the major considerations made during preparation of off site emergency plan for a construction site.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the various types of Scaffoldings (with diagram) used at construction sites. List the safety provisions to be considered during erection of scaffoldings at construction site.
- Q.37 What are the notifiable diseases? Explain the cause, symptoms, and preventions for of the following:
- Silicosis
 - Asbestosis
 - Toxic nephritis
 - Occupational dermatitis
 - Caissons disease
- Q.38 Describe the various Hazards linked to tunneling works and safety provisions to be taken before and during execution of the same.

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Roll No.

1st Year / Industrial Safety

Subject:- Safety in Construction Industries

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the permissible limit of exposure-time weighed exposure (TWA 8hrs) for hydrogen sulphide?
- 10 ppm
 - 5 ppm
 - 15 ppm
 - None of the above
- Q.2 Vertical members in a trestle scaffold is called
- Standards
 - Ledger
 - Transom
 - Bracing
- Q.3 As per the BOCW (RE&CS) Act, 1996, the appropriate government in case of CPSU (Central Public Sector Undertaking) Company is
- State government of concerned state where it is located
 - Central Government
 - Both (a) and (b)
 - None
- Q.4 The type of water retaining structures of facilitate underground construction, which are permanent in nature becomes part of final structure is called
- Cofferdams
 - Caisson
 - Piling
 - Pedestal
- Q.5 Which of the following is not considered as Building & Other Construction work as per section 2 (d) of BOCW (RE&CS) Act, 1996?
- Construction works in area notified under the Factories Act, 1948
 - Construction work related to railways

- c) Construction works in area notified under the Mines Act, 1952
 d) Both a) & c)
- Q.6 Which one of the following is a not part of scaffold and is used for providing anchorage:
 a) Ledger b) Bracings
 c) Lifeline d) Standards
- Q.7 The use of pneumatic tools and chain saw cause:
 a) Whole body vibration
 b) Hand-arm vibration
 c) Both a and b
 d) None of the above
- Q.8 While providing ladder for exit from trench, the ladder should project at least ____m from top level of excavated trench as per IS 3674?
 a) 2 b) 1.25
 c) 0.75 d) 1
- Q.9 The following is indirect cost of accident
 a) Money paid for treatment of worker
 b) Compensation paid to worker
 c) Cost of lost time of injured worker
 d) All of the above
- Q.10 Which of the following require training and authorization before you can perform them?
 a) Forklift operation b) Crane operation
 c) Electrical Works
 d) Housekeeping works
 e) Scaffolding erection works

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What do you mean by Flame proof equipment?
 Q.12 What do you mean by strutting in case of excavation works?

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- Q.13 What do you mean by caisson disease?
 Q.14 What do you mean by Well Sinking?
 Q.15 What do you mean by Horizontal and Vertical Lifelines in case of Working at Height Jobs?
 Q.16 Give full form of ELCB?
 Q.17 What do you mean by LOTO procedure in context of electrical works?
 Q.18 What do you mean Dangerous Occurrence?
 Q.19 What do you mean by Root Cause Analysis in Accident Investigation at a construction site?
 Q.20 What is the function of an anemometer in case of a tower crane operation?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a short note on safety provisions to be taken during excavation at a construction site.
 Q.22 List various types of excavation equipment used at construction sites with their use case.
 Q.23 List and describe various PPEs used during Working at Height Jobs.
 Q.24 Write a short note on provisions related to first aid facilities stipulated under BOCW (RE&CS) Act, 1996 and Rules framed there under.
 Q.25 List down the welfare provisions under BOCW (RE&CS) Act, 1996 and Rules frames there under.
 Q.26 Write a note on safety provisions to be followed while handling and storage of paints and varnishes at construction sites.
 Q.27 Write a note on safety provisions to be followed while carrying out scaffolding works at a construction sites.
 Q.28 Why do we require shoring?
 Q.29 Explain the safety precautions to be followed during hot works like welding, grinding etc at a construction site.

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